

ABHISHEK KUMAR

+44-7393068156 • abhishekkumar0598uk@gmail.com • www.linkedin.com/in/abhishek-kumar-a0190a18b • United Kingdom

Education

The University of Manchester

United Kingdom

Master of Science in Data Science - Distinction (Achieved)

09/2023 - 09/2024

- **Relevant Coursework:** Statistics (Linear Algebra) and Machine Learning, Understanding Databases, Programming in Python for BA, Understanding Data & its Environment, Financial Data Analytics and AI in Finance, Text Mining.

SRM Institute of Science and Technology

India

Bachelor of Technology in Information Technology - Distinction (Achieved)

05/2016 - 05/2020

- **Relevant Coursework:** Python, R Programming, Data Science and Big Data Analytics, DBMS, Data Mining, Data Warehousing, Statistics.

Experience

Data Scientist - Finance | MQube

03/2024 - 09/2024

- Developed production-grade **ML pipelines** for financial market prediction using **AWS SageMaker** and **Lambda**, aligned with **MLOps** best practices.
- **Fine-tuned** domain-specific LLMs for **sentiment analysis** and returns **forecasting**, enhancing risk-adjusted returns by 30%.
- Used **quantitative modeling** and rolling statistical windows to develop adaptive ML pipelines responsive to real-time financial trends.
- Communicated model performance, risk metrics, and trade-offs clearly to both technical and business stakeholders.

Data Scientist - Geospatial | Equans

11/2023 - 02/2024

- Analyzed Scope 3 emissions across multi-modal travel data (air, train, car, accommodation) using **Databricks** and **MySQL** for data exploration and ETL, with **ArcGIS** and **Power BI** for geospatial analysis and dashboarding, optimizing routes to reduce CO₂ emissions by 75% and costs by 20%.
- Developed automated **ETL** pipelines and applied **geospatial analysis** (Shapely, GeoPandas, ArcGIS, NetworkX) to map high-emission routes, align corporate travel policies with sustainability goals, and drive data-driven decision-making for business decision-makers.

Data Scientist - Fraud and Risk | National Payments Corporation of India

11/2020 - 08/2023

- Developed **30+** real-time and batch fraud detection models using **Random Forest**, **Autoencoders**, and **XGBoost**, reducing fraud by 35% and false positives by 20%.
- Built end-to-end ML pipelines on **Azure**, using **Azure Data Factory** for ETL, **Azure ML** for model training/versioning, and **AKS** for deployments.
- Designed high-throughput data ingestion workflows using **Kafka** and **Azure Event Hubs**, processing over 2 billion UPI transactions monthly.
- Built a **churn prediction model** using user activity patterns, transaction frequency, and engagement drop-offs to forecast user retention risk.
- Collaborated cross-functionally with compliance, risk, and backend teams to align model logic with requirements and real-time system constraints.
- Led **A/B testing** and model evaluation cycles, building **KPI-driven Power BI dashboards** using **Power Query** to help decision-makers interpret model performance and trade-offs.
- Maintained production/dev grade codebases, authored technical documentation, and supported integration for 300+ enterprise clients.

Projects

DocuChat-RAG

- Built an **Agentic RAG chatbot** using CrewAI, LangChain, Qdrant, and Streamlit for Q&A on PDFs and web search.
- Integrated local **LLM inference (Ollama)**, **prompt engineering**, and **fallback search (FireCrawl)** for dynamic knowledge retrieval.

TextScout: Customer Transcript Analysis Using NLP and Generative AI

- Built an **NLP** pipeline using **RoBERTa**, **BART (HuggingFace)**, **FAISS**, and **prompt engineering** for real-time **sentiment** and **zero-shot classification**.
- Implemented self-learning AI with automated retraining triggered by **FAISS**, improving model classification accuracy.

Fraud Detection in Credit Card Transactions

- Built an **unsupervised anomaly detection** pipeline using **Python**, **NumPy**, **Scikit-Learn**, and unsupervised learning methods for fraud detection.
- Engineered robust feature scaling and **PCA** for **dimensionality reduction**, achieving 84% recall while minimizing false positives.

NLP-Driven Relation Extraction with Advanced Neural Networks

- Developed a deep learning model combining **BiLSTM**, **CNN**, and **Multi-Head Attention** to extract semantic relationships from text.
- Leveraged **GloVe embeddings**, **NLTK** for NLP preprocessing and **TensorFlow/Keras** to achieve a 75% F1 score on SemEval-2010 Task 8.

DiaRisk: Clinical Risk Assessment for Diabetes

- Developed empirical models using **logistic regression** and **tree-based methods** to assess health risks using patient-level datasets.
- Owned the full ML pipeline: from **EDA** and model experimentation, designing **PowerBI** dashboard to **MLOps** integration using **Docker** and **AWS**.

Skills

Technical Skills:

Python (Pandas • NumPy • Matplotlib • Seaborn • PyTorch • OOP • scikit-learn) • TensorFlow • R • Java • PowerBI • DAX • Advanced SQL • PostgreSQL • MongoDB • DBT • Big Query • Machine Learning • Decision Trees • Random Forests • Time Series Forecasting • Predictive modeling • Anomaly Detection • Deep Learning • Neural Networks • Natural Language Processing (NLP) • LLMs • RAG • Prompt Engineering • LangChain • CrewAI • AWS (Sagemaker • S3 • Lambda • Quicksight) • Microsoft Azure • Databricks • Version Control (Git) • Linux and Bash • Docker • MLOps • MLFlow • CI/CD • FastAPI.